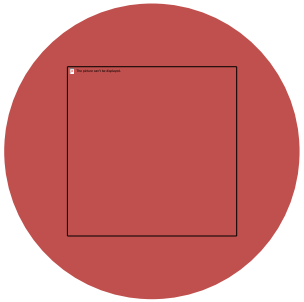
The Art of Subtle Automation: Getting It Done Without Scope Creep

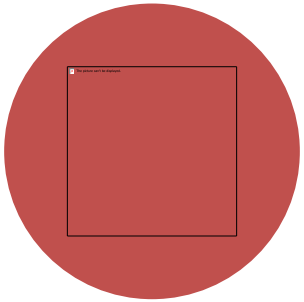
How to Embed Test Automation Without Derailing Delivery

Olha Vovk

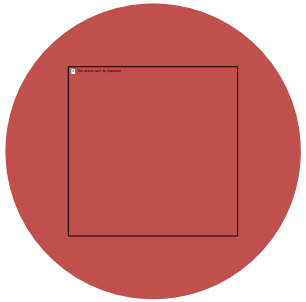
What this talk is and isn’t



NOT A HOW-TO FOR A SINGLE TOOL

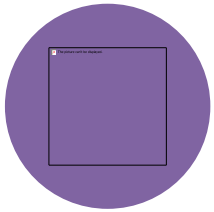


IT'S A MINDSET AND PRACTICAL TOOLKIT

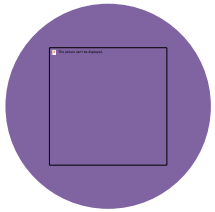


NOT THE BEST PRACTICES GUIDE

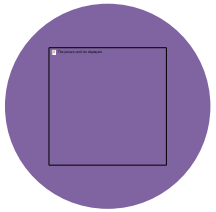
Agenda



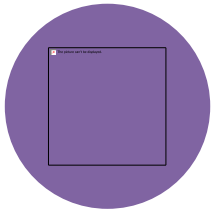
PROBLEM



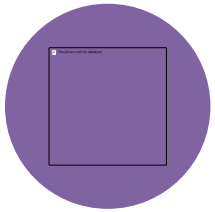
SOLUTION



TOOLS

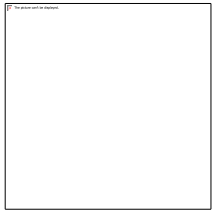


STORIES FROM THE TRENCHES

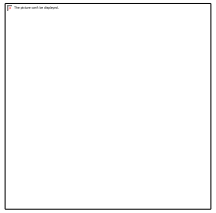


TAKEAWAYS

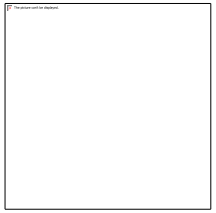
Problem Statement



- Automation is often pushed out of scope



- Time-to-market pressures dominate delivery timelines



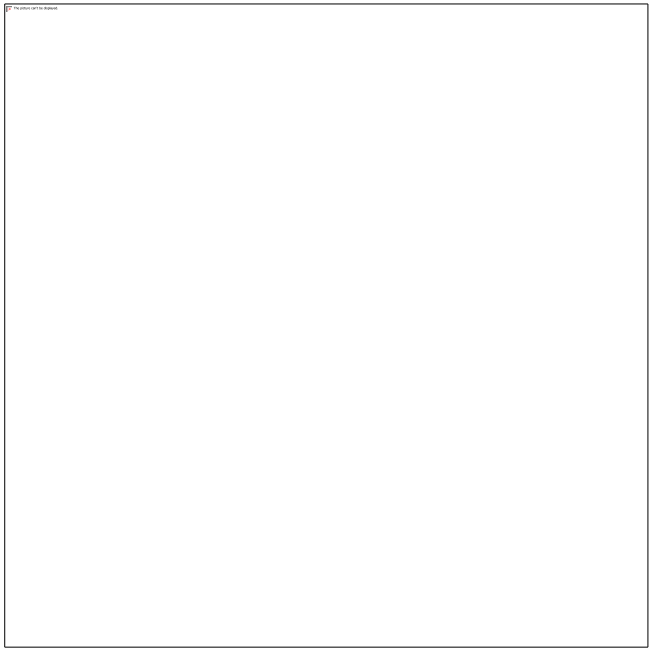
- Business sees test automation as overhead

THE STORY THAT BROUGHT US HERE

The story of what brought me from architecting automation solutions for complex systems to having to sneak in automation into the process

Consequences of Ignoring Automation

- Mounting technical debt
- Slower feedback loops long term
- Manual regression fatigue
- Reinforcement of unscalable practices



SOLUTION

Time to act

The Mindset Shift

Test automation is not about WHAT we do, but HOW we do TESTING

"Automate without asking for permission"

Also, without additional budget

What is Subtle Automation?

Subtle automation enhances manual testing by embedding small, helpful scripts, utilities, or plugins that reduce repeat effort and increase speed and/or precision without requiring a total coverage.

Subtle Automation Will Not

The picture can't be displayed.

Give you comprehensive test coverage

The picture can't be displayed.

Might not be scalable

The picture can't be displayed.

Might not be a part of CI/CD

Subtle Automation Will

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Speed up your testing

Provide you with early issue detecting mechanism

Save your sanity during fast release cycles

Strategy that worked for my team

- Unit testing included in the dev DoD
- Set up automation tools, early, while outsourcing test case/plan creation to AI
- Start automating what's possible now
- Prioritize high-impact scripts
- Collaborate with devs, DevOps, Data Analytics
- Bring Observability to Test environment
- Leverage AI tools
- Get comfortable with imperfect automation solutions
- Don't mention *test automation maintenance* in presence of Product owner

TOOLS

Helpful tools for test automation

Tool 1: Postman

- Quick API validation
- Ease of test automation
- Collections + tests = fast coverage
- Chaining requests into business flows
- CLI for easy CI/CD integration
- No excuse not to automate

API Network

Search Postman Ctrl K

Invite

Upgrade

Import

Overview

GET get screenshot status ba

AWS-STAGE

QA Test for DMM Device Actions API V1 / Tests / **get screenshot status bad**

Share

GET https:// {{domainName}} /dmm/v1/action/screenshot/UNKNOW

Send

Params Authorization Headers (11) Body Scripts Settings

Cookies

Pre-request

Post-response

```
pm.test("Status code is 400", function ()
    pm.response.to.have.status(400);
});

pm.test("Test the response", () => {
  var jsonData = JSON.parse(pm.response.te
  pm.expect(jsonData).to.be.an("object");
  pm.expect(jsonData.status).to.exist;
  pm.expect(jsonData.status).to.equal(400)
  pm.expect(jsonData.message).to.exist;
  pm.expect(jsonData.message).to.contain("UNKNOWN")
});
```

Search snippets

Tests

- Status code: Code is 200
- Response body: Contains string
- Response body: JSON value check
- Response body: Is equal to a string
- Response headers: Content-Type header check
- Response time is less than 200ms
- Status code: Successful POST request

Packages Snippets

s API V1

py path

py path

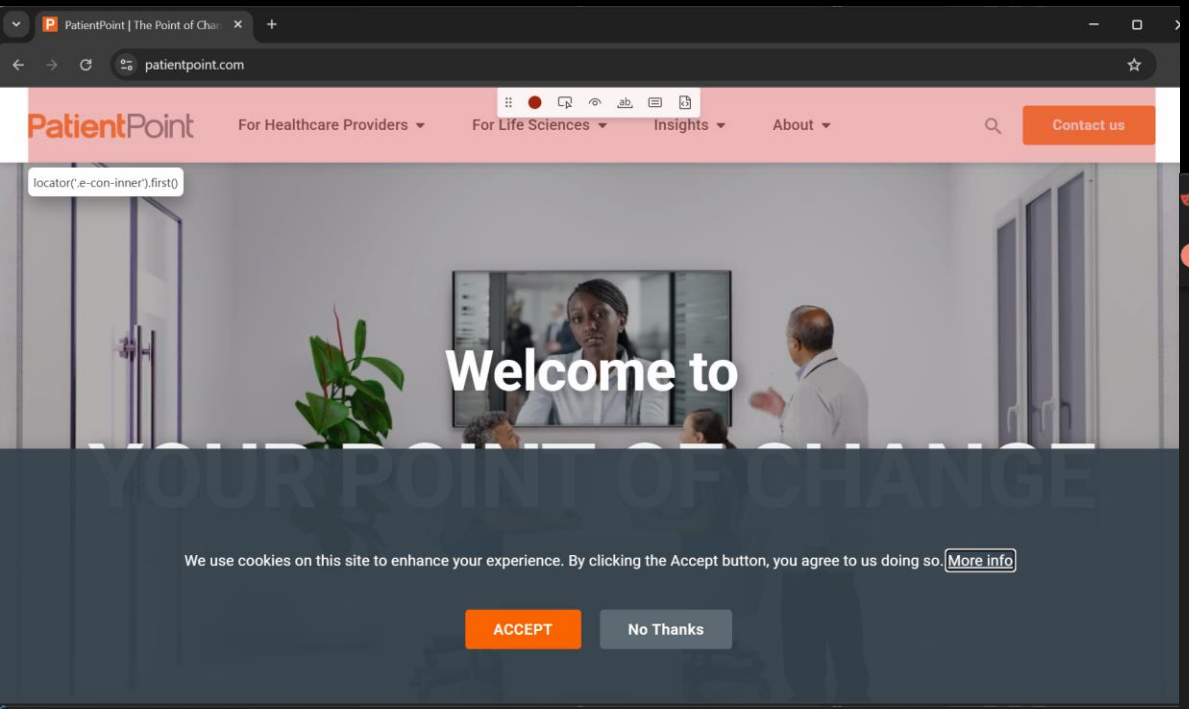
n happy ...

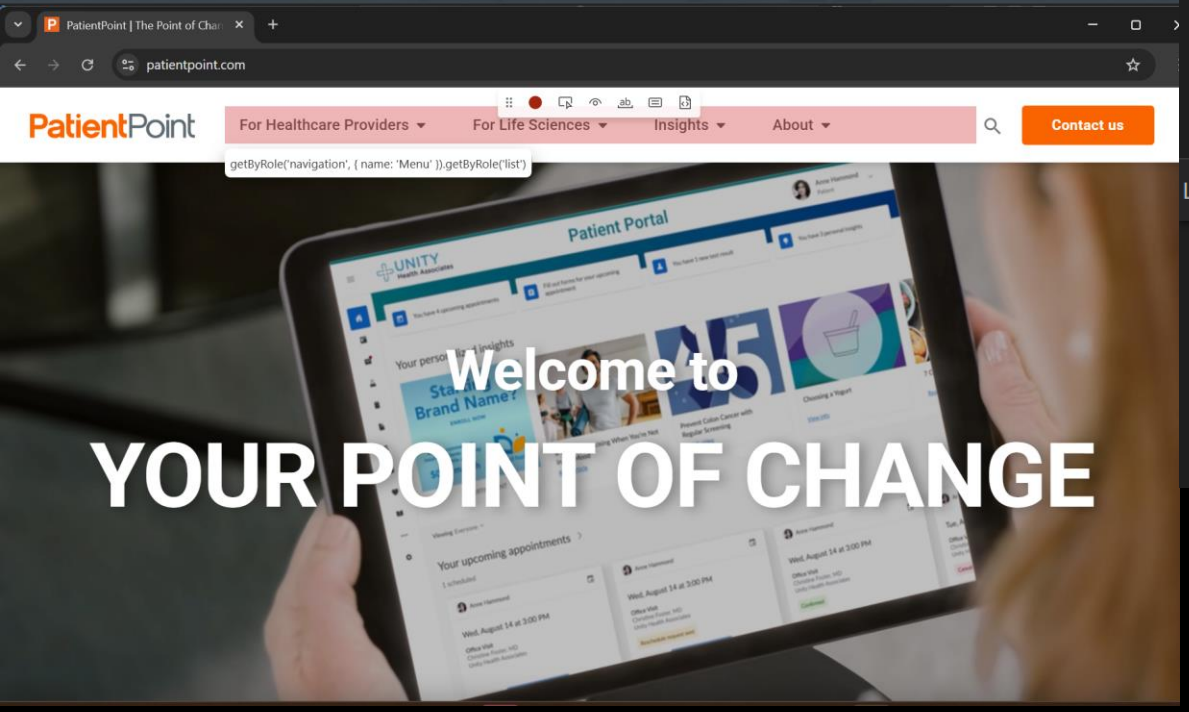
happy p...

happy p...

Tool 2: Playwright

- Fast browser automation
- Popular unit test framework
- E2E test framework
- Works well in CI/CD pipelines
- VS Code extension for record/play
- Short tests, high value
- Pause for user input with **page.pause()**





Playwright Inspector - https://patientpoint.com/

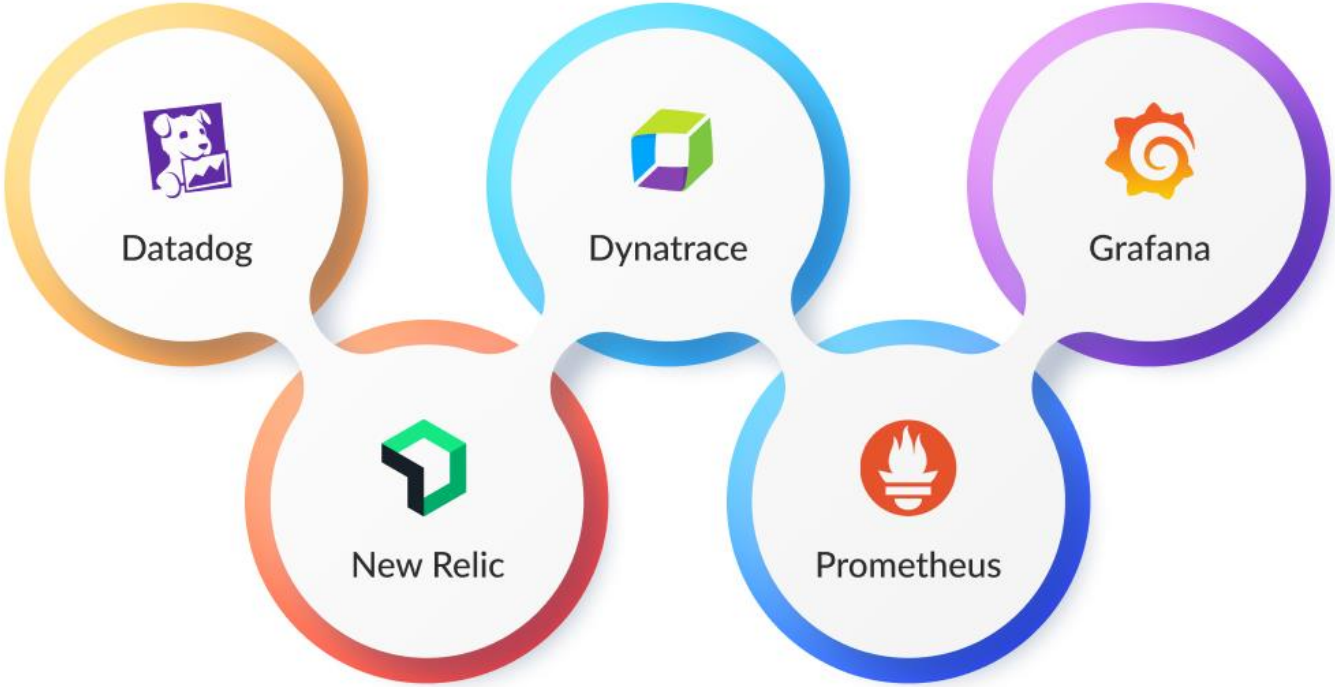
Record | Tar | Test

```
1 import { test, expect } from '@playwright/test';
2
3 test('test', async ({ page }) => {
4   await page.goto('https://patientpoint.com/');
5   await page.getByRole('link', { name: 'No Thanks' 
6 });
```

Locator Log Aria

Tool 3: Observability in Test Environment

- Set up observability tool in test environment
- Create alerts on abnormal behaviors
- Detect issues without writing tests



Tool 4: Cross-IDE Testing Toolkit

Tool	Category	Primary Use	Works With
SonarLint	Code Quality & Security	Detects bugs, vulnerabilities, and code smells in real time	VS Code, IntelliJ, PyCharm, Eclipse
Snyk	Security & Dependency Scanning	Finds vulnerabilities in dependencies, containers, IaC	VS Code, JetBrains IDEs, Eclipse
TestNG	Unit & Integration Testing (Java)	Flexible Java testing framework for various test types	IntelliJ, Eclipse, VS Code
Pytest	Python Testing	Unit, API, and functional testing with fixtures & plugins	VS Code, PyCharm, CLI
ESLint	Linting & Code Style (JS/TS)	Prevents logic/style errors that cause test failures	VS Code, JetBrains IDEs, Sublime
Jest	Unit & Integration Testing (JS/TS)	Seamlessly run JavaScript/TypeScript tests and see results inline for faster TDD cycles.	VS Code, Eclipse, JetBrains IDEs, Atom, Sublime

Tool 5: Generative AI

Tool	Free Plan Details	Primary Use Case
ChatGPT	Free access with limited GPT-4	Manual & automated test-case ideas
GitHub Copilot	Free for students & open-source devs	Inline test generation in IDE
TestGPT	Free tier available	Generate cases from user stories
CodiumAI	Free community version	Generate meaningful unit tests
TestRigor	Free plan with limits	No-code automation from plain English
QA Wolf	Free tier with limited runs	Playwright automation in plain English
Katalon TestOps AI	Free plan for small teams	AI-assisted test design & maintenance

Other Tools

- BackstopJS – for visual test automation
- WebLinkValidator or similar - for dead links validation
- Chrome Lighthouse audits (has CI/CD integration)
- Screaming Frog (SEO tools) for spell checking and more
- Data Analytics tools (Tableau, Power BI) for data validation

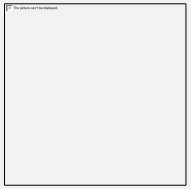
WHAT OTHER NON-TEST AUTOMATION TOOLS HAVE YOU USED FOR TESTING?

STORIES FROM THE TRENCHES

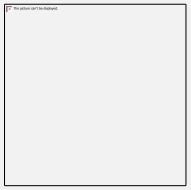
Real Story: Negative Login Tests

- Login functionality wasn’t automation friendly
- Tester automated Playwright negative tests
- Automated positive test with the Pause for user entry

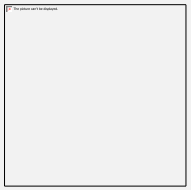
The Gain



Confidence for each release cycle



Built foundation and experience for automation



Creating the “muscle memory”

Real Story: Postman for Data Setup

- Functionality had the need for data setup
- Creating data was taking over 30 min of manual work
- Used APIs for creating test data
- Postman run the API with various data payloads
- Fast data cleanup with a Postman collection

The Gain

Test data setup took ~ 2 hours in Postman once and seconds to run

Test coverage for the filtering functionality has increased for each cycle dramatically

Increased reliability and speed of release cycle

Created a test suite for a different flow detecting API issues before testing UI

Real Story: GitHub Copilot Test Creation

- Prompted Copilot to generate test cases for functionality
- Cleaned up test cases
- Prompted to create automated scripts for those tests
- Ran with multiple data sets

The Gain

- Repeatable automated test execution
- Can test multiple data scenarios
- Saved time on creating the manual test case
- Building experience with AI tools

Real Story: Utility Scripts for Data Validation

- Event logs generated by IoT devices reported data in json format and pushed in real time
- Data aggregated into large files in the storage service
- Python utility script created for data filtering from the aggregated file

The Gain

Saved the team 10+ hours of tedious manual labor per cycle

Increased precision of data validation

Increased reliability and speed of release cycle

Utility can be later re-used in automated tests

Real Story: Visual tests for educational portal

- Used robots.txt to get all URLs
- Added all URLs to the BackstopJS visual test script
- Ran once to grab all the baseline images
- Ran to visually verify all the pages
- Executed every major release to ensure visual integrity

The Gain

Saved the team hours of tedious manual labor per cycle

Increased CSS test coverage for various viewports

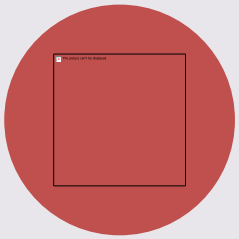
Increased reliability and speed of release cycle

Team can reliably experiment with changing styles and design

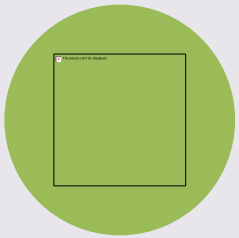
WHAT STORIES DO YOU HAVE TO SHARE?

No Judgement Zone Here

Framework for Action

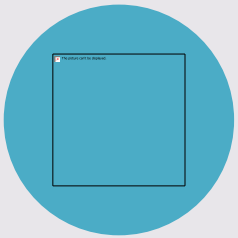


BRING THE TESTING WITH AUTOMATION TOOLS MINDSET TO THE TEAM



AUTOMATE FOR EFFICIENCY TODAY AND NOT FOR AUTOMATION BEST PRACTICES

USE CURRENT TOOLS BETTER (OBSERVABILITY)



START USING AI TOOLS NOW TO GAIN EXPERIENCE AND TRAIN THE MODEL ON YOUR DATA

Final Takeaways

- Subtle automation is doable
- Automate now, don’t ask for permission
- Use your full arsenal of tools
- Reach out for help

Thank You / Stay in touch!

@olga-vovk-qa-engineer